

Analyzing the Relationship Between Trading Volume, Market Conditions, Exchange Rate Movements, and Stock Returns

Project Proposal

Introduction

Financial markets are highly dynamic systems affected by both firm-level activity and broader macroeconomic conditions. Investors continuously monitor variables such as trading volume, market trends, and exchange rate fluctuations in order to better understand stock price behavior and make investment decisions.

In emerging markets such as Turkey, exchange rate volatility and market uncertainty can significantly influence stock returns across different sectors. Therefore, examining the combined effect of market conditions, trading activity, and macroeconomic indicators may provide valuable insight into stock market behavior.

This project aims to analyze the relationship between trading volume, overall market conditions, exchange rate movements, and stock returns using real-world financial datasets from Borsa Istanbul.

In addition to statistical analysis, machine learning techniques will also be applied to investigate whether stock return movements can be predicted using historical financial indicators and macroeconomic variables.

Objective of the Project

The primary objective of this project is to investigate whether trading volume, market conditions, and exchange rate movements have a statistically significant effect on stock returns.

More specifically, the project aims to:

- Analyze the relationship between stock returns and trading activity
- Compare stock behavior with overall market performance using the BIST100 index
- Examine how USD/TRY exchange rate fluctuations affect stock returns
- Measure volatility and trend behavior across different sectors
- Apply machine learning techniques to predict stock return direction
- Evaluate whether predictive performance differs between sectors

The project combines financial analysis, statistical methods, and machine learning approaches to provide a comprehensive examination of stock market behavior.

Datasets and Data Collection

The datasets used in this project will be collected programmatically from Yahoo Finance using Python libraries such as `yfinance`, `pandas`, and `numpy`.

The analysis will focus on real-world financial datasets obtained from Borsa Istanbul.

Dataset 1 – Stock Price Data

Daily stock market data will be collected for selected companies traded on Borsa Istanbul.

Selected Stocks

- THYAO.IS
- GARAN.IS
- AKBNK.IS
- ASELS.IS
- SISE.IS
- EREGL.IS

These companies were selected because they represent different sectors of the Turkish economy, allowing both sectoral comparison and broader market analysis.

Variables Collected

- Open price
- Close price
- High price
- Low price
- Trading volume
- Adjusted close price

Time Range

- 2018–2025

The dataset is expected to contain approximately:

- 250 trading days per year
- Around 2,000 observations per stock
- Approximately 12,000 total observations

Using multiple stocks from different sectors improves the generalizability of the analysis and allows comparison of sector-specific behavior.

Dataset 2 – Market Index Data

Daily BIST100 index (XU100) data will be used to represent overall market conditions.

The market index will serve as a benchmark for evaluating whether individual stock returns move consistently with overall market behavior.

Using BIST100 instead of average stock returns provides a more realistic and economically meaningful representation of market conditions.

Dataset 3 – Exchange Rate Data

USD/TRY exchange rate data ($USDTRY=X$) will be incorporated into the analysis to capture macroeconomic effects on stock returns.

Exchange rate fluctuations are particularly important in Turkey because they may influence:

- Investor sentiment
- Inflation expectations
- Import/export costs
- Corporate profitability
- Market volatility

Including exchange rate movements strengthens the macroeconomic dimension of the project and allows a more comprehensive analysis of stock return behavior.

Variables Used in the Analysis

The project will include both firm-level and macroeconomic variables.

Main Variables

- Daily stock returns
- Trading volume
- Market returns (BIST100 returns)

- Exchange rate returns
- Lagged returns
- Rolling volatility
- Rolling averages
- Stock price levels

Feature Engineering

Additional variables will be generated during preprocessing to improve the analytical and predictive capability of the project.

Examples include:

- Previous-day returns
- Percentage changes
- Moving averages
- Volatility indicators
- Lagged trading volume variables

These engineered variables may help capture temporal relationships and improve machine learning performance.

Research Questions

The project aims to answer the following research questions:

- Is there a statistically significant relationship between trading volume and stock returns?
- How strongly do market conditions influence stock return behavior?
- Do exchange rate movements affect stock returns across sectors?
- Which sectors are more sensitive to market volatility and exchange rate changes?
- Can machine learning models successfully predict stock return direction?
- Which financial indicators contribute most to prediction performance?

These questions aim to connect statistical analysis with practical financial interpretation.

Hypotheses

Null Hypothesis (H0)

Trading volume, market conditions, and exchange rate movements do not have a statistically significant effect on stock returns.

Alternative Hypothesis (H1)

Trading volume, market conditions, and exchange rate movements have a statistically significant effect on stock returns.

The hypothesis testing stage will evaluate whether the observed relationships are meaningful or occur randomly.

Methodology

The project will consist of several analytical stages.

1. Data Collection and Preprocessing

Financial datasets will be collected automatically using Python APIs.

Preprocessing Steps

- Handling missing values
- Date formatting
- Calculating daily returns
- Normalizing data
- Generating lagged variables
- Creating volatility indicators

Data preprocessing is essential to ensure consistency and reliability before conducting statistical and machine learning analyses.

2. Exploratory Data Analysis (EDA)

EDA will be conducted to better understand stock market behavior and identify patterns within the datasets.

Analyses Performed

- Distribution analysis
- Correlation analysis
- Volatility analysis
- Trend analysis

- Sector comparison
- Cumulative return analysis

Visualizations Used

- Histograms
- Boxplots
- Correlation heatmaps
- Line charts
- Return comparison graphs
- Volatility plots

EDA will help identify relationships between trading volume, market movements, exchange rate fluctuations, and stock returns.

3. Statistical Analysis

Statistical methods will be applied to examine relationships between variables.

The analysis will focus on:

- Correlation between trading volume and stock returns
- Relationship between market returns and individual stock performance
- Impact of exchange rate changes on stock returns
- Volatility behavior across sectors

Hypothesis testing will be used to determine whether the relationships are statistically significant.

4. Machine Learning Analysis

Machine learning classification models will be implemented to predict whether the next day's stock return will be positive or negative.

Machine Learning Model

- Logistic Regression

Features Used

- Trading volume
- Lagged returns
- Market returns
- Exchange rate movements
- Volatility indicators

ML Workflow

1. Feature selection
2. Train-test split
3. Model training
4. Prediction generation
5. Performance evaluation
6. Result interpretation

The machine learning stage will evaluate whether historical financial indicators can provide predictive insight into stock return behavior.

Expected Contribution

This project aims to contribute to financial data analysis by combining:

- Firm-level financial indicators
- Market-wide variables
- Macroeconomic variables
- Machine learning techniques

The study is expected to provide insight into how trading activity, market conditions, and exchange rate fluctuations influence stock returns in Turkey.

In addition, comparing multiple sectors may help identify whether certain industries are more sensitive to macroeconomic and market-related changes.

The project also demonstrates how data science techniques can be applied to real-world financial datasets to generate meaningful analytical and predictive insights.

Technologies and Libraries Used

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- yfinance

Platforms

- Google Colab
- GitHub

Conclusion

This project combines financial analysis, statistical methods, and machine learning techniques to investigate stock market behavior in Turkey.

By integrating trading volume, market conditions, exchange rate movements, and predictive modeling into a single analytical framework, the project aims to provide a more comprehensive understanding of stock return dynamics and financial market behavior across different sectors.